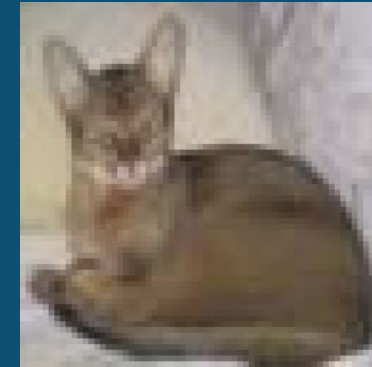


SELF-SUPERVISED

Image Super-Resolution

4x single-image super-resolution, trained without a premade LR/HR dataset — pairs synthesized from ordinary photos.



LR INPUT



MODEL OUTPUT

Filling in the gaps, beyond simple stretching

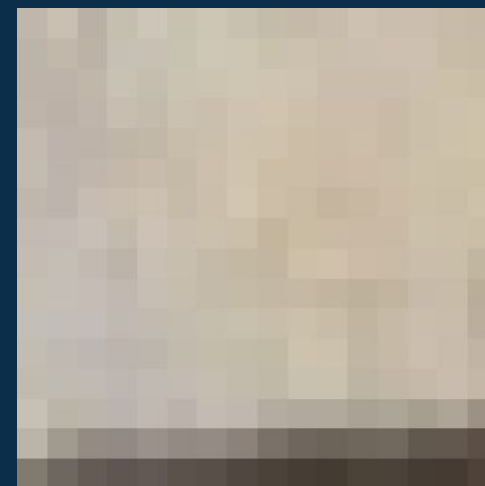
You know how when you zoom into a photo too far, it gets blocky and blurry? Super-resolution is the task of generating a bigger, sharper version of it — not just stretching it, but filling in detail that looks like it belongs there.

More than just “resizing”

Bicubic interpolation (standard resizing) is mathematically safe, but it simply smears existing pixels. Super-resolution tries to invent plausible new detail that isn't in the source image at all.

An irreversible process

Going from low-res to high-res is ill-posed: real detail was permanently discarded, so there's no single “correct” answer — only plausible ones, like guessing a cat's exact fur texture from a thumbnail.



This is a photo, zoomed in until the pixels show

Every image is a grid of these tiny colored squares — a pixel. More pixels = higher resolution.

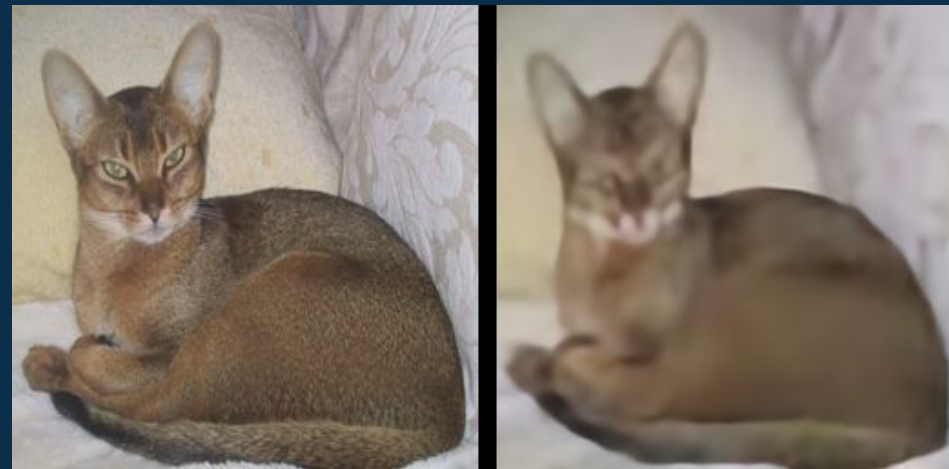
The real question: can a model guess detail that generalizes to real, messy photos — not just clean textbook examples?

One low-res image, many valid answers

Super-resolution is ill-posed: a single low-resolution image is consistent with countless plausible high-resolution originals. The model isn't recovering the “true” image — it's learning which reconstruction is most plausible.

4x

upscale factor — 16x more pixels
to infer per image, none of them
directly observed

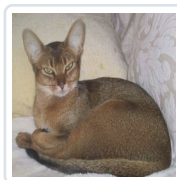


Ground truth vs. bicubic-degraded input

Bicubic interpolation is the mathematically “safe” answer — and it's blurry. Learned models exist to do better than safe.

No premade dataset — pairs are manufactured

1



Source pool

Oxford-IIIT Pet photos used only as a raw HR image pool — never sourced as a paired SR dataset.

2

HR stays HR

Source images are never upsampled. Only kept if at or above a minimum native resolution.

3

LR is synthesized

Every low-res training input is generated on the fly from an HR photo via a randomized degradation pipeline.

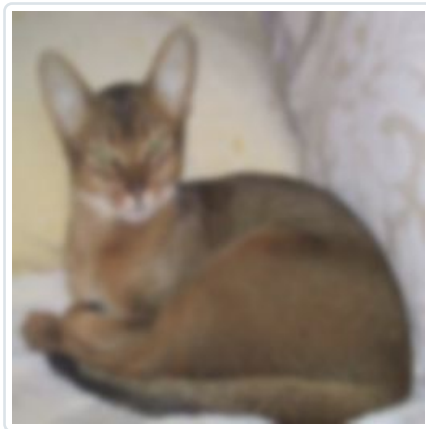
This is the “self-supervised” part: no human-labeled LR/HR pairs are downloaded. The supervision signal comes entirely from HR photos plus a synthesis process the model never sees the parameters of.

Four stages, randomized per sample

1

Blur

Simulates lens softness / motion



after step 1

2

Downsample

Bicubic, 4x — the actual loss



after step 2

3

Noise

Sensor noise, low-light capture



after step 3

4

JPEG re-compress

Real-world storage artifacts



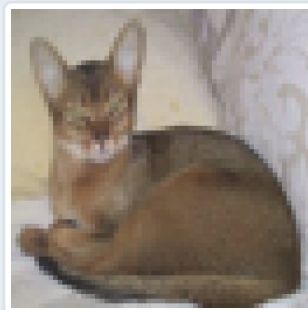
after step 4

Every parameter — blur radius, noise level, JPEG quality — is randomized independently for every training sample, not fixed per epoch. This is one possible random draw, shown cumulatively.

Why randomized degradation, not clean bicubic?

The naive approach

Train on bicubic-downsampled inputs only. Clean, simple — but the model learns to invert one specific, deterministic operation, and overfits to “undoing bicubic resizing” specifically.



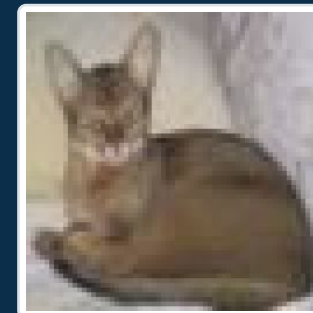
bicubic-only input

The failure mode

Real low-quality photos are blurry, noisy, and JPEG-compressed all at once — the model never learned to handle that combination, so it generalizes poorly.

This project's approach

Randomize blur, noise, and JPEG quality independently per sample — matching BSRGAN / Real-ESRGAN-style degradation modeling. The model sees a much wider distribution of “bad” images, so it learns features that hold up under many kinds of degradation, not just one.



full-pipeline input

This is exactly what the ablation on slide 10 tests directly.

Three models, increasing complexity



same real photo, same crop

1

Bicubic

NON-LEARNED FLOOR

Classical interpolation. No training, no learned parameters.



same real photo, same crop

2

SRCNN

DONG ET AL.

9x9 $\rightarrow$ 1x1 $\rightarrow$ 5x5 conv stack. Simplest learned baseline.



same real photo, same crop

3

SRResNet

FINAL MODEL

Residual blocks + PixelShuffle + no batch norm
+ delta prediction.

SRResNet: four ideas, four papers

1

Residual blocks

Ledig et al. — SRResNet

Backbone of stacked residual blocks for feature extraction at low resolution, before any upsampling.

2

PixelShuffle upsampling

Shi et al. — ESPCN

Learned upsampling staged in sequential 2x steps rather than one 4x jump — sharper, fewer artifacts.

3

No batch normalization

Lim et al. — EDSR

Normalization was found to hurt SR quality by disrupting the pixel intensity information the model needs to preserve.

4

Residual / delta prediction

Kim et al. — VDSR

The network predicts a correction on top of a bicubic upsample, not the full image from scratch.



delta (exaggerated)

Loss function: L1 + a touch of perceptual

Why not MSE?

MSE's quadratic penalty biases the model toward blurry “safe average” predictions on an ill-posed problem — exactly the failure mode SR needs to avoid.

L1 as the primary term

Linear penalty — doesn't over-punish small pixel errors the way MSE does, so predictions stay sharper.

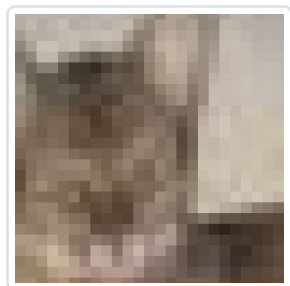
+ VGG16 perceptual term

weight = 0.01 — added on top of L1 to counteract residual blur that pure-pixel losses leave behind.

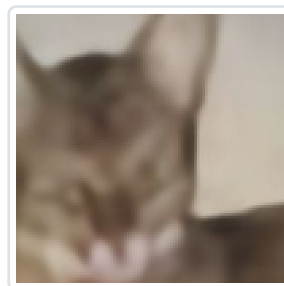
Deliberately not implemented

Full adversarial (GAN) loss — skipped due to training instability and a mismatch with PSNR/SSIM-based evaluation.

Same detail region, side by side:



LR input (blurry)



model output (sharp)

The whole point of L1 + perceptual over MSE: keep detail like this instead of averaging it away.

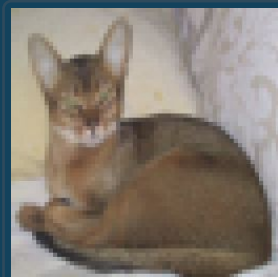
The ablation: does realistic degradation matter?

Same architecture. Same loss. One variable changes.

RUN A — BICUBIC-ONLY

Zero blur / noise / JPEG

Trained only on clean bicubic-downsampled inputs — the naive approach from slide 6.

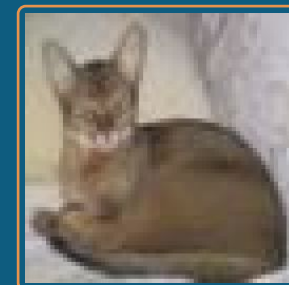


VS

RUN B — FULL DEGRADATION

The randomized pipeline

Trained on blur + downsample + noise + JPEG, fully randomized per sample.



The question: would a model that never saw noisy, blurry training data still perform well on real low-quality photos? The results table and real-photo grid answer this directly.

PSNR / SSIM on the held-out test split

Model	PSNR	SSIM
Bicubic	24.780	0.6456
SRCNN	24.945	0.6528
SRResNet (bicubic-only)	24.966	0.6467
SRResNet (full degradation)	25.614	0.6750

+0.84

dB PSNR

gap between full-degradation and bicubic-only
SRResNet — the headline ablation result.

Bicubic < SRCNN < SRResNet-bicubic < SRResNet-full



Reminder: this is what “winning by 0.84 dB” looks like — not huge in isolation, but consistent across the whole test set.

Real-photo comparison, no ground truth



Columns: LR input → bicubic → SRCNN → SRResNet-bicubic → SRResNet-full

Why this matters more than the table

These are genuinely low-quality photos — old scans, screenshots, compressed downloads — with no ground-truth HR version. They were never used in training.

This is where the ablation's claim gets tested for real: does the full-degradation model actually generalize, or did it just win on a synthetic metric?

Future work

1

Adversarial (GAN) loss

Trade some PSNR/SSIM for sharper, more texture-realistic output — the most substantive omission.

2

Attention mechanisms

Channel or spatial attention to let the model weight informative regions more heavily.

3

Blind degradation estimation

Predict degradation parameters at inference time instead of assuming the training distribution.

Thank you

github.com/CMB-i/self-supervised-sr